

Team Contributions

Project: Semantic Paper Search (RAG Application)
Course: BANA 275 - Distributed Computing / Analytics
Date: February 2026

1. [Ziqi(Peter) Huang]

Application Architecture: Designed the overall RAG (Retrieval-Augmented Generation) flow and selected the tech stack (Streamlit + OpenAI).

UI Implementation: Built the ``paper_doc.py`` frontend using Streamlit, including the sidebar configuration and results display grid.

Search Engine Logic: Implemented the ``cosine_similarity`` search function and top-k ranking algorithms.

Integration: Handled the OpenAI API connection for both Embeddings (``text-embedding-3-large``) and Chat Completion (``gpt-4o``).

2. [Member 2 Jui Yu(Raynee) Kuo]

Data Ingestion: Implemented the ``parse_pdf`` function using ``pypdf`` to extract raw text from uploaded files.

Tokenization Strategy: Developed the Sliding Window Chunking algorithm (in ``chunk_text``) to solve the 8,192 token limit issue, ensuring large papers are processed correctly.

Debugging: Identified and fixed critical bugs related to DataFrame handling and data types during the embedding process.

Deployment: Managed the Python environment and created the ``requirements.txt`` to ensure reproducibility.

3. [Member 3 Xuanzhu Luo]

Documentation: Authored the ``README.md`` and ``ARCHITECTURE.md``, explaining the technical design and setup instructions.

Presentation: Created the project demo video, including script writing and voiceover narration.

User Experience: Refined the user interface messages (success/error toasts) to improve usability during the "Processing" phase.

Repo Management: Organized the GitHub repository structure and managed branch merges.

4. [Member 4 Min Hsuan(Ivan) Lai]

Prompt Engineering: Tuned the system prompt in ``summarize_results`` to ensure the AI cites sources and strictly adheres to the provided context.

ArXiv Integration: Implemented the alternative data source option to fetch papers dynamically via the ArXiv API.

Validation: Conducted "hallucination tests" by asking known questions against specific papers to verify the accuracy of the semantic search.

Code Ownership

`paper\_doc.py` (Core Logic): [Ziqi Huang] & [Min Hsuan Lai]

`requirements.txt`: [Jui Yu Kuo]

Documentation Files: [Xuanzhu Luo]

Statement of Collaboration

We confirm that this project was a collaborative effort. While members took ownership of specific technical components, all team members participated in the initial design planning and final code review.